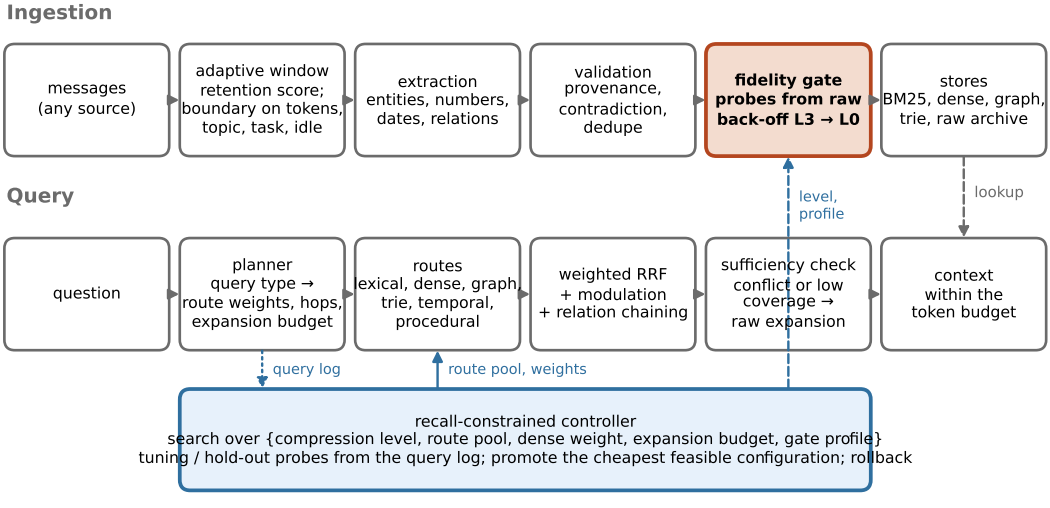
Ingestion
messages (any source)
adaptive window retention score; boundary on tokens, topic, task, idle
extraction entities, numbers, dates, relations
validation provenance, contradiction, dedupe
fidelity gate probes from raw back-off L3 → L0
stores BM25, dense, graph, trie, raw archive
Query
level, profile
lookup
question
planner query type → route weights, hops, expansion budget
routes lexical, dense, graph trie, temporal, procedural
weighted RRF + modulation + relation chaining
sufficiency check conflict or low coverage → raw expansion
context within the token budget
query log
route pool, weights
recall-constrained controller
search over {compression level, route pool, dense weight, expansion budget, gate profile}
tuning / hold-out probes from the query log; promote the cheapest feasible configuration; rollback